

CONTENT DECLINE RANKING

Final Model Development and Validation Report

Human-in-the-loop ranking of pages at risk of future search-impression decline

61,795

final modeling pages

74.6%

mean Precision@100

+39.4 pp

vs Week-4 rule

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github.com/Dania-Yasir/flyrak-project

Executive Summary

Version note

This document supersedes the earlier working report supplied during model development. Historical experiments are retained only when they explain a design decision. Where an older number differs from the current repository, the current deterministic rerun is treated as authoritative.

This project asks whether search-performance information available in the first half of a month can rank pages that later experience a material decline in Google Search impressions. The output is a review queue, not an automated content decision.

Using the March 2026 partition of the FlyRank ML Internship warehouse, the final outcome-evaluable cohort contains 61,795 pseudonymized pages across 34 clients. Features use March 1-15 only; March 16-31 is reserved for the future-decline proxy.

The selected Random Forest uses 21 pre-outcome features covering momentum, performance level, trend shape, and client-relative context. The final evaluation holds out complete clients in five balanced grouped folds.

Under the deterministic client-disjoint rerun, the full model reached 74.6% mean Precision@100 (SD 13.6%), compared with 66.2% for a momentum-only Random Forest and 35.2% for the Week-4 rule baseline evaluated on the same held-out folds.

The evidence supports a human-in-the-loop ranking workflow: model prioritizes, human investigates, human decides. It does not establish causal refresh impact, predict Google's ranking algorithm, or guarantee equal performance across clients or future months.

Key final evidence

Measure	Final value	Interpretation
Overall decline rate	32.4%	Base rate in the final March cohort
Full-signal RF P@100	74.6% +/- 13.6%	Primary client-disjoint result
Momentum-only RF P@100	66.2% +/- 12.8%	Stronger learned control
Week-4 rule P@100	35.2% +/- 6.4%	Fair same-fold baseline
Full vs Week-4	+39.4 pp	Observed improvement on the same folds
Full vs momentum-only	+8.4 pp	Directional gain from added context

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Acknowledgments and data credit

1. Project Question and Decision Context

Research question: Can search-performance signals available in the first half of a month help prioritize which content pages should be reviewed first for possible future decline?

The practical problem is review capacity. Large content portfolios can contain many pages, while analysts can inspect only a subset. The goal is therefore to rank pages so that higher-risk candidates appear earlier in a human review queue.

Operating principle

Model prioritizes -> human investigates -> human decides.

- Unit of analysis: one pseudonymized content page within one pseudonymized client.
- Output: a risk score and ranked review queue.
- Decision supported: which pages should an analyst investigate first when review capacity is limited.
- Not supported: automatic rewriting, deletion, publishing, or claims about the causal effect of a refresh.

Outcome proxy

Component	Definition
Feature window	March 1-15, 2026
Outcome window	March 16-31, 2026
Decline proxy	Average daily outcome-window impressions are more than 20% lower than average daily feature-window impressions
Interpretation	Decision-support proxy for future impression decline; not a universal definition of content decay

2. Data, Cohort, and Information Boundary

The analysis uses the FlyRank ML Internship warehouse release, build v20260703, accessed through DuckDB and Hugging Face. The primary table is fact_content_daily_performance, which contains approximately 78.8 million daily performance rows across the full warehouse release.

Stage	Pages	Clients
Raw aggregated March page rows	331,437	55
Feature-eligible before future filtering	68,288	35
Final outcome-evaluable modeling cohort	61,795	34

The final cohort has an observed future-decline rate of 32.4%. The reduction from the aggregated March population is material: the model applies to pages with complete Google Search Console coverage in the required

feature and outcome windows. Results should therefore be interpreted for this eligible population, not automatically for every page in the warehouse.

Information boundary and leakage control

The prediction boundary is the end of March 15. Every model input must be available by that point. The outcome period is used only to construct the target.

- client_id and content_id are used only for grouping and validation, never as model inputs.
- Future impressions, future average daily impressions, impression_change_pct, and is_declining_proxy are excluded from features.
- Client-relative percentile features are calculated from pre-outcome data only.
- The final audit found no forbidden field overlap, no suspicious future-derived feature names, and no missing values in the final 21-feature matrix.
- Public outputs use pseudonymized, aggregate information only; no client names, URLs, or private queries are included.

3. Transparent Week-4 Baseline

The Week-4 baseline was deliberately simple. It represents a transparent rule an analyst could understand without a learned model.

1. A page must have average first-half search position of 20 or better.
2. Its CTR must be low relative to the median CTR for pages in the same search-position bucket.
3. Matching pages are ranked by first-half impressions so higher-visibility opportunities appear first.

The position buckets are Top 3, 4-10, 11-20, 21-50, and 50+. The action rule is restricted to Top-20 pages because the earlier signal audit showed that the relationship between low CTR and later decline weakened outside the stronger visibility ranges.

Why two baseline numbers exist

The original Week-4 Precision@100 was 39.0% when the rule ranked the full March queue once. The fair final comparison is 35.2% because the same rule is re-evaluated inside the exact client-disjoint folds used by the learned models. The two numbers answer different evaluation questions.

4. Model Design

The selected model is a Random Forest classifier using 21 pre-outcome features. A momentum-only Random Forest using two impression-momentum features is retained as a stronger learned control so that the value of the additional context can be tested directly.

Feature group	Count	What it captures
Momentum anchors	2	Recent impression movement
Performance level	5	Impressions, clicks, CTR, average position, activity rate
Trend shape	10	Five-day steps, acceleration, CTR/position/activity movement, recent trajectory
Client-relative context	4	Within-client relative traffic, CTR,

		position, and recent trend
Total	21	All available before the prediction boundary

Final Random Forest settings

Parameter	Value
n_estimators	200
max_depth	10
min_samples_leaf	25
max_features	sqrt
class_weight	balanced_subsample
random_state	42
n_jobs	-1

The final settings are frozen before evaluation. This avoids selecting hyperparameters on held-out evaluation folds.

5. Validation Strategy and Methodological Lessons

The most important methodological lesson was that random row-level validation was too optimistic for this problem. Pages from the same client can share portfolio-specific patterns, so mixing one client across training and validation makes the task easier than the intended deployment setting.

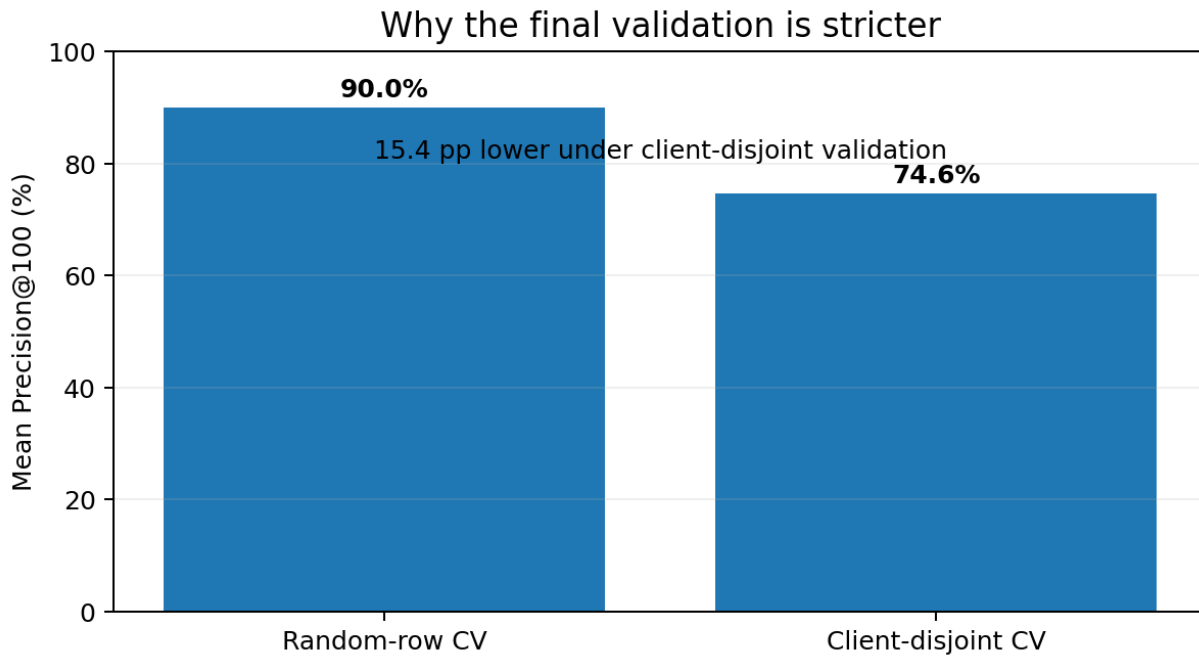


Figure 1. Random-row validation versus client-disjoint validation.

Takeaway: Holding out complete clients reduces the optimistic estimate from 90.0% to 74.6% Precision@100, a 15.4 percentage-point drop.

In the final random-row cross-validation, about 29.6 clients were present on both the training and validation sides on average. The client-disjoint design removes this overlap: complete clients are held out from training in each fold.

Why earlier grouped designs were not enough

The earlier development report documented several grouped validation designs. They were useful for exposing cross-client difficulty, but some produced outer folds with very uneven client counts, including one-client folds. Those experiments are retained as development evidence, not as the final estimator.

The final primary evaluation uses five balanced client groups, approximately 7 / 7 / 7 / 7 / 6 clients. This makes each fold more interpretable while preserving the client-disjoint requirement.

6. Final Deterministic Results

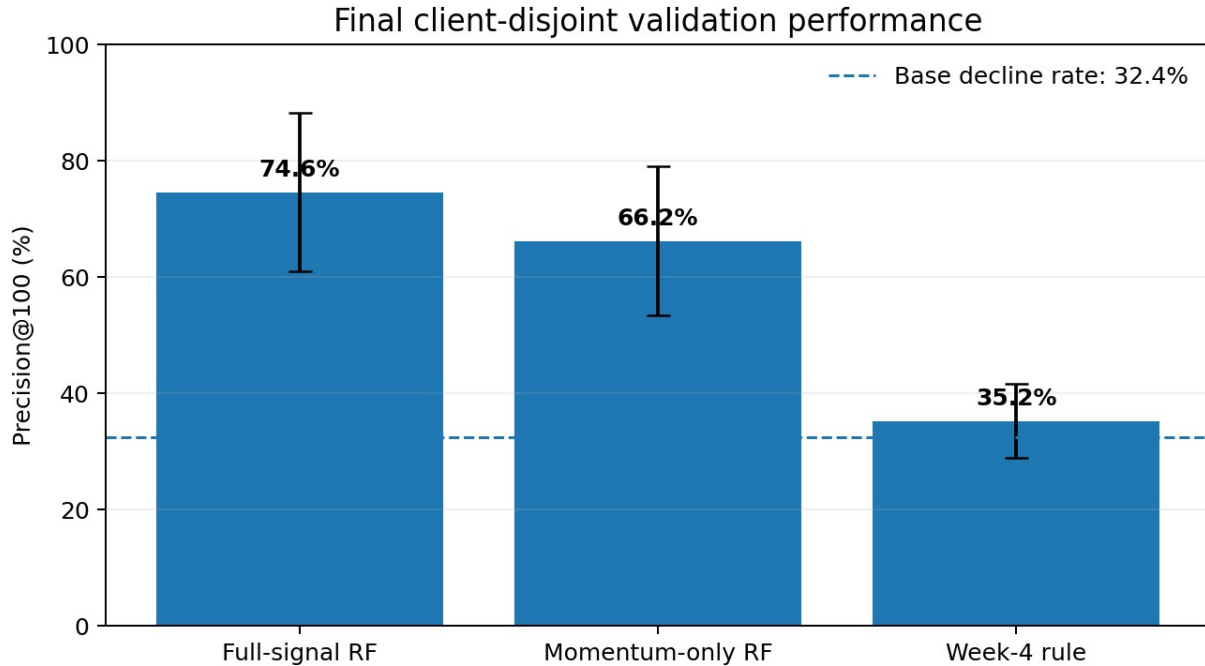


Figure 2. Final primary validation performance with fold-to-fold standard deviation.

Takeaway: The full-signal Random Forest is materially stronger than the Week-4 rule and shows a smaller directional gain over the momentum-only Random Forest.

Approach	Mean P@100	SD	Difference vs full
Random Forest - Full Signal	74.6%	13.6%	-
Random Forest - Momentum Only	66.2%	12.8%	-8.4 pp
Week-4 Rule Baseline	35.2%	6.4%	-39.4 pp

The 74.6% result is the primary number for the project because it uses the final deterministic row order and the client-disjoint evaluation design. A previous working report contained nearby values such as 75.8%; those older values are superseded by the deterministic rerun.

The 8.4-point difference over the momentum-only Random Forest is more informative about feature value than the much larger gap over the rule baseline. It suggests that trend shape, performance level, CTR, position, and client-relative context add useful ranking information beyond recent momentum alone, while fold variability remains substantial.

Secondary client-level metric

A separate macro-client Precision@10 check treats each eligible client more equally. It is a different metric and should not be combined numerically with fold-level Precision@100.

Approach	Macro-client P@10
Random Forest - Full Signal	69.2%
Random Forest - Momentum Only	58.8%
Week-4 Rule Baseline	38.8%

7. Ranking Utility Under Limited Review Capacity

Precision@100 measures the quality of the very top of a queue. A complementary operational question is how much of the observed decline population is captured when an analyst reviews a fixed share of ranked pages.

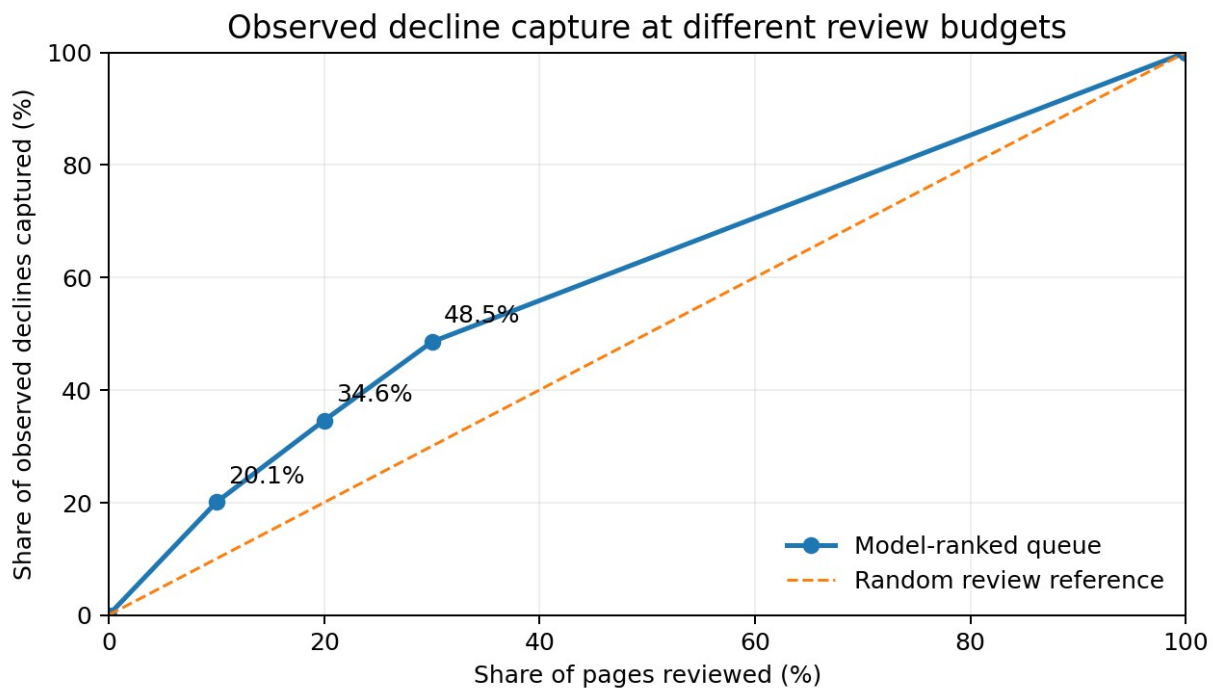


Figure 3. Cumulative decline capture at different review budgets.

Takeaway: Reviewing the top 20% of the out-of-fold queue captured 34.6% of observed decline cases; the ranking therefore concentrates more decline cases near the top than a random review order.

Review budget	Observed declines captured
Top 10% of pages	20.1%
Top 20% of pages	34.6%
Top 30% of pages	48.5%

These capture rates support the ranking use case, but they are not estimates of business impact or causal value. They describe where observed decline cases appear in the out-of-fold queue.

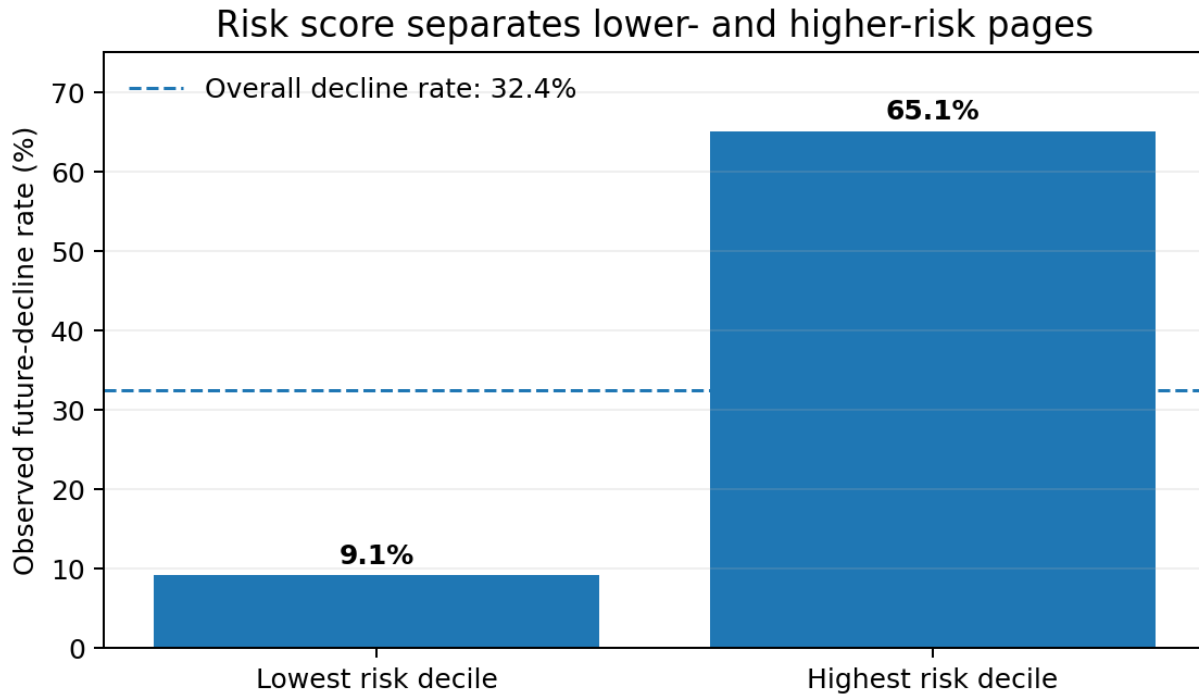


Figure 4. Observed decline rate in the lowest and highest model-risk deciles.

Takeaway: The highest risk decile has a 65.1% observed decline rate versus 9.1% in the lowest decile, showing useful score ordering.

8. Cross-Client Generalization

Pooled metrics can be influenced by large clients. The final action notebook therefore also checks whether high-risk pages have a higher observed decline rate than each client's own overall decline rate.

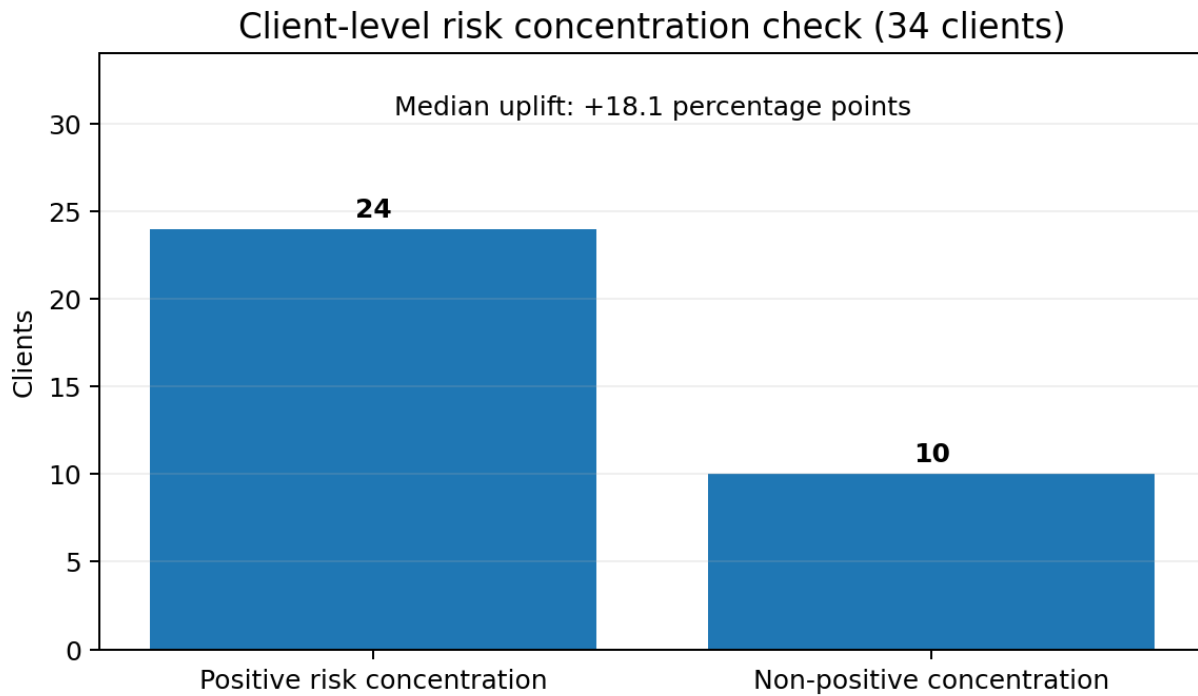


Figure 5. Client-level risk concentration summary.

Takeaway: Across 34 evaluated clients, 24 showed positive concentration of decline in the high-risk group, with a median uplift of about +18.1 percentage points.

This result is encouraging but deliberately not described as universal. Ten clients did not show positive concentration in this check, and the 13.6% fold-level standard deviation also shows material client heterogeneity.

Deployment implication

A global model can be useful as a common ranking layer, but client-level monitoring remains necessary. Performance should be re-checked when a client portfolio, traffic regime, or measurement coverage differs materially from the March 2026 evaluation cohort.

9. Model Interpretation and Error Analysis

Feature interpretation

The earlier development report included a Random Forest feature-importance audit. The useful qualitative finding remains that the model relied on more than the original momentum signal: recent-vs-prior impression trend, the latest five-day impression step, client-relative trend context, traffic level, position, CTR, and momentum all appeared among the stronger inputs.

These importances are used only as a model-behavior sanity check. Random Forest importance does not establish that a feature causes future decline.

Out-of-fold error patterns

The prior report also documented qualitative error patterns from predictions made while each page's client was held out. These observations remain useful for understanding the task, even though they are not treated as final headline metrics.

Error type	Typical pre-outcome pattern	Interpretation
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High-ranked false positive	Negative momentum, multiple declining steps, worsening position	The warning looked real by March 15 but did not persist enough to cross the later -20% label threshold
Low-ranked true decline	Flat or positive first-half momentum, little decline shape, sometimes improving position	The later reversal left little warning signal before the prediction boundary

This error structure is important: the model can rank evidence that is present before March 15, but it cannot reliably anticipate a future reversal that leaves little or no pre-outcome trace.

10. Action Playbook

The final model score is converted into an operational review policy. The percentile thresholds are capacity rules, not learned causal cutoffs.

Risk percentile	Queue state	Default handling
≥ 0.80	Active human review	Assign a review type using observable reason codes
0.50 to < 0.80	Watchlist	No immediate edit; monitor and review as capacity allows
< 0.50	Monitor	No immediate edit

High-risk review types

Observed pattern	Recommended action
Sustained decline pattern	CONTENT_REFRESH_REVIEW
Ranking slippage	SERP_AND_INTENT_REVIEW
Visible page with relatively low CTR	TITLE_META_CTR_REVIEW
Client-relative anomaly	MANUAL_DIAGNOSTIC_REVIEW
High risk without a clean heuristic explanation	HUMAN_REVIEW_BEFORE_EDIT

No-go rule

A recommendation is an investigation priority, not an instruction to rewrite, publish, delete, redirect, deindex, change canonical tags, or alter URL structure automatically. Reason codes describe observable patterns; they are not causal explanations.

11. Limitations and Next Validation Steps

- Proxy outcome: the target measures future impression decline, not content quality and not the causal benefit of an intervention.
- Coverage selection: only 61,795 of 331,437 aggregated March page rows enter the final modeling cohort because complete GSC coverage is required.
- Client count: the final cohort contains 34 clients. Client-disjoint validation is necessary but still based on a limited number of groups.
- Unequal client sizes: some clients contribute far more pages than others, which contributes to fold-level variability.
- Client heterogeneity: performance is not equally strong across every client or fold.
- Portfolio context assumption: client-relative features require enough same-client pages at scoring time to compute comparable percentiles.
- Single primary month: March 2026 is the main modeling and validation period. Temporal transfer is not established by the final headline result.
- No causal interpretation: feature importance, risk scores, and reason codes describe associations useful for ranking; they do not explain why Google Search performance changed.

Recommended next validation work

4. Run a frozen temporal holdout on a later month without re-selecting the model on that month.
5. Repeat client-disjoint evaluation across additional periods to test drift and seasonality.
6. Track analyst review outcomes so model ranking quality can be separated from intervention effectiveness.
7. If model tuning is reopened, use nested or otherwise isolated tuning inside training data so evaluation folds remain untouched.
8. Monitor client-level Precision@K and risk concentration rather than relying only on a pooled global metric.

12. Reproducibility

The repository is the reproducibility source of record. The main notebooks map the project from data definition through action policy.

Notebook	Role
work/notebooks/w03_data_contract.ipynb	Defines windows, cohort, target, and leakage boundary
work/notebooks/w04_baseline_score.ipynb	Builds and evaluates the transparent Week-4 rule
work/notebooks/w05_model_training!.ipynb	Model development experiments
work/notebooks/w06_validation_audit.ipynb	Final deterministic validation and claim audit
work/notebooks/w07_action_playbook.ipynb	Final queue, review policy, figures, and paper metrics
work/notebooks/capstone.ipynb	Concise capstone summary that mirrors the deployed paper

Environment and fixed choices

- Python with pandas, NumPy, scikit-learn, matplotlib, DuckDB, and huggingface_hub.
- Hugging Face token is supplied through Colab Secrets or getpass; it is never committed.
- Random seed: 42.

- Primary evaluation: five client-disjoint folds.
- Feature window: March 1-15, 2026.
- Outcome window: March 16-31, 2026.
- Dataset: FlyRank ML Internship warehouse, build v20260703.

Repository: <https://github.com/Dania-Yasir/flyrak-project>

Dataset page: <https://huggingface.co/datasets/FlyRank/internship-warehouse>

13. Final Project Statement

Selected model

Random Forest - Full Signal

The final evidence supports a client-disjoint, human-in-the-loop ranking system for prioritizing pages at higher risk of future search-impression decline.

The strongest defensible result is not a universal accuracy claim. Under the final deterministic March 2026 evaluation, the full model achieved 74.6% mean Precision@100, materially above the Week-4 rule and directionally above a momentum-only Random Forest, while performance varied substantially across held-out client groups.

The project should therefore be presented as decision support: it concentrates observed decline cases near the top of a review queue and gives analysts structured reason codes for investigation. It should not be presented as a prediction of Google's ranking algorithm or as proof that refreshing a page causes recovery.

Appendix A. Historical Development Evidence Retained from the Earlier Report

The supplied 12-page working report captured useful experiments that pre-date the final deterministic rerun. The following findings are retained because they explain why the final design looks the way it does. They are historical development evidence, not the final headline metrics.

Development finding	Earlier evidence	Why it mattered
Random-row validation was optimistic	Examples included 80.8% vs 63.0% and 89.8% vs 75.8% random vs grouped P@100	Motivated client-disjoint evaluation
Some grouped folds were poorly balanced	An earlier outer design had client counts 1 / 1 / 8 / 11 / 13	One-client folds were not accepted as the primary estimator
Stability-only feature removal weakened the advantage	A stricter cohort produced RF 69.0%, Logistic 68.8%, Momentum 68.0%	Showed that removing portfolio context also removed useful signal
Repeated optimized balancing was too repetitive	40 seeds produced only four unique optimized partitions in the earlier stress audit	Prevented an overly optimistic stability claim

These older results are intentionally separated from the final main tables. Current repository values - especially 74.6%, 66.2%, and 35.2% - supersede nearby earlier numbers when reporting the final model.

Appendix B. Final Feature and Model Configuration

21 final pre-outcome features

Group	Feature
Momentum	imp_momentum_pct
Momentum	imp_momentum_log
Level	log_imp_first_half
Level	log_clicks_first_half
Level	ctr_first_half
Level	avg_position_first_half
Level	active_rate_first_half
Trend shape	imp_step12_log
Trend shape	imp_step23_log
Trend shape	imp_acceleration
Trend shape	imp_declining_steps
Trend shape	imp_recent_vs_prior_log
Trend shape	click_step23_log
Trend shape	ctr_change_5d_pp
Trend shape	position_change_5d
Trend shape	active_rate_change_5d
Trend shape	has_position_shape
Client context	log_imp_client_percentile
Client context	ctr_client_percentile
Client context	position_client_percentile
Client context	recent_vs_prior_client_percentile

Excluded outcome-related fields

- client_id

- content_id
- imp_future
- avg_daily_imp_future
- impression_change_pct
- is_declining_proxy

Acknowledgments and Data Credit

This capstone was completed using the FlyRank ML Internship dataset and project framework. The analysis uses the approved internship warehouse release and follows the repository's public-safety rules for pseudonymized data, leakage control, and cautious claim language.

FlyRank: <https://flyrank.com/>

FlyRank ML Internship warehouse: <https://huggingface.co/datasets/FlyRank/internship-warehouse>

Project repository: <https://github.com/Dania-Yasir/flyrak-project>

Source reconciliation note: an earlier working report supplied during model development was used for historical context, especially validation-design experiments, splitter sensitivity, feature interpretation, and qualitative error analysis. Final metric values in this document are aligned to the current repository and deterministic rerun.